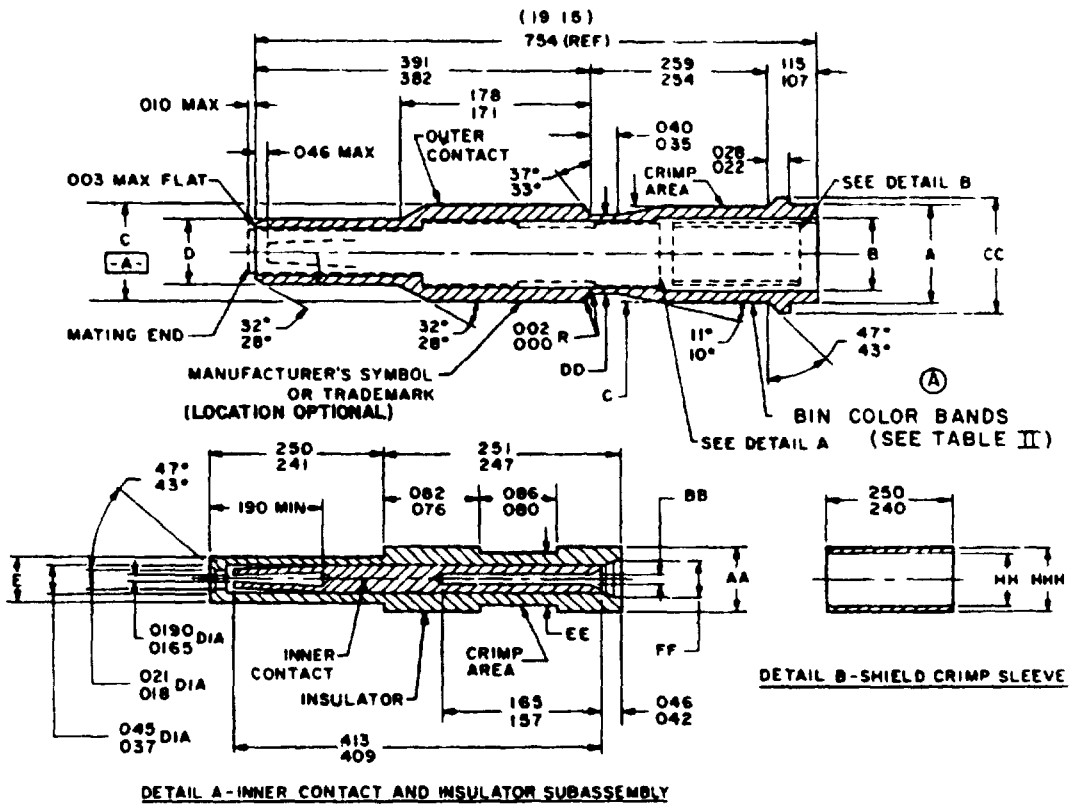
MILITARY SPECIFICATION SHEET

# CONTACTS, ELECTRICAL CONNECTOR, PIN, CRIMP REMOVABLE, SHIELDED, (FOR MIL-C-81511 SERIES 1 AND 2 CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the contacts described herein shall consist of this document and the latest issue of Specification MIL-C-39029.

INACTIVE FOR NEW DESIGN




DETAIL B-SHIELD CRIMP SLEEVE

DETAIL A-INNER CONTACT AND INSULATOR SUBASSEMBLY

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses for overall length and diameter only.
4. Dimensions shown apply after plating.

FIGURE 1. Connector contact.



FSC 5935

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TABLE I. Dimensions.

| BIN code | A DIA | B DIA | C DIA | D DIA | E MAX DIA | AA DIA | BB MIN DIA | CC DIA | DD DIA | EE DIA | FF MIN DIA | HH MIN DIA | HHH MAX DIA |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 120 | .103 | .0910 | .113 (2.87) | .071 | .058 | .081 | .022 | .133 | .099 | .068 | .037 | .058 | .087 |
| | .101 | .0885 | .110 (2.79) | .069 | | .078 | | .130 | .096 | .062 | | | |
| 121 | .162 | .1430 | .161 (4.09) | .1165 | .093 | .128 | .022 | .190 | .145 | .076 | .067 | .086 | .130 |
| | .159 | .1405 | .158 (4.01) | .1140 | | .126 | | .187 | .142 | .070 | | | |
| 122 | .162 | .1430 | .161 (4.09) | .1165 | .093 | .128 | .022 | .190 | .145 | .076 | .108 | .110 | .130 |
| | .159 | .1405 | .158 (4.01) | .1140 | | .126 | | .187 | .142 | .070 | | | |
| 123 | .162 | .1430 | .161 (4.09) | .1165 | .093 | .128 | .022 | .190 | .145 | .076 | .108 | .128 | .142 |
| | 159 | .1405 | .158 (4.01) | .1140 | | .126 | | .187 | .142 | .070 | | | |
| 124 | 162 | .1430 | .161 (4.09) | .1165 | .093 | .128 | .034 | .190 | .145 | .089 | .098 | .128 | .142 |
| | .159 | .1405 | .158 (4.01) | .1140 | | .126 | | .187 | .142 | .085 | | | |
| 125 | .162 | .1430 | .161 (4.09) | .1165 | .093 | .128 | .034 | .190 | .145 | .089 | .098 | .110 | .130 |
| | 159 | .1405 | .158 (4.01) | .1140 | | .126 | | .187 | .142 | .085 | | | |



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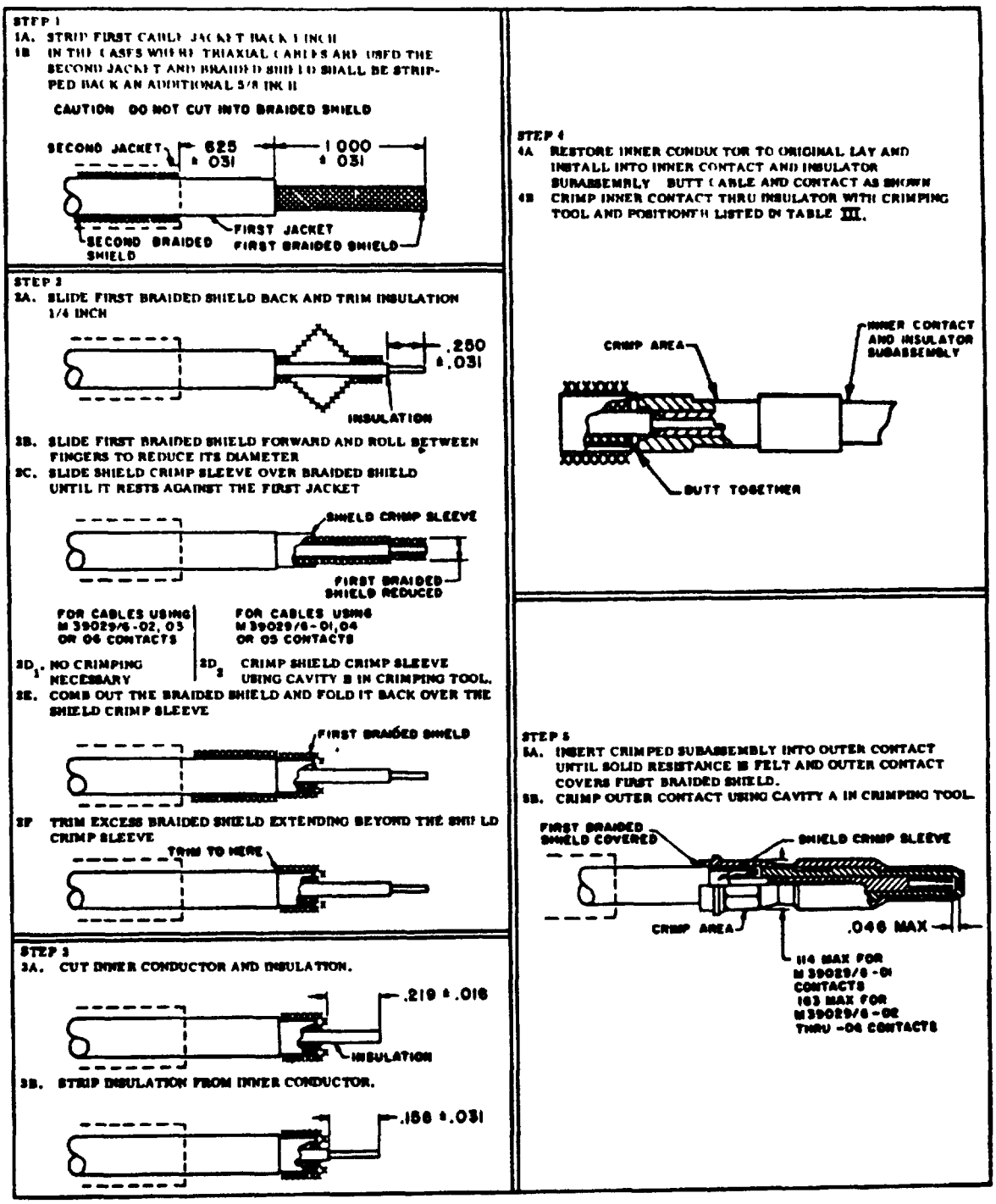



FIGURE 2. Assembly procedure.



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TABLE II Design characteristics.

| BIN code | Color bands | | | Contact cavity size | Cables accommodated | Type | Class |
|---|---|---|---|---|---|---|---|
| | 1st | 2nd | 3rd | | | | |
| 120 | Brown | Red | Black | 16 | RG-178A/U<br>RG-196A/U | D | B |
| 121 | Brown | Red | Brown | 12 | RG-179B/U<br>RG-187A/U<br>RG-188A/U | | |
| 122 | Brown | Red | Red | 12 | 9530D5117<br>(RAYCHEM) 1/ | | |
| 123 | Brown | Red | Orange | 12 | RG-180B/U<br>RG-195A/U<br>293-3922<br>(MICRODOT) 1/ | | |
| 124 | Brown | Red | Yellow | 12 | 250-4070<br>(MICRODOT) 1/ | | |
| 125 | Brown | Red | Green | 12 | 5022E5111<br>(RAYCHEM) 1/ | | |

1/ or equivalent

REQUIREMENTS.

Dimensions, design characteristics, and configuration See figure 1 and tables I and II.

Tools See table III

Mating contact MIL-C-39029/13 and MIL-C-39029/14

Manufacturer's recommended assembly instructions to be shipped with unit package.

Assembly procedure See figure 2.

Preparation of samples. Contacts shall be wired as required using wire in accordance with table II (cables accommodated column).

Contact resistance Contact resistance at a load current of 1.0 ampere shall meet the requirements of table IV.

Contact engagement and separation forces The contact separation and engagement forces shall meet the requirements of table IV.

Dielectric withstanding voltage 1,000 volts, ac, rms from sea level to 110,000 feet altitude

Tensile strength The tensile strength shall meet the requirements of table IV.

QPL evaluating activity Naval Weapons Support Center, Code 3074, Crane, Indiana 47522



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TABLE III Tools.

| BIN code | Inner contact | | Shield crimp sleeve | | Outer contact | | Installing tool | Removal tool |
|---|---|---|---|---|---|---|---|---|
| | Basic crimping tool | Positioner | Basic crimping tool | Die | Basic crimping tool | Die | | |
| 120 | M22520/2-01 | M22520/2-26 | M22520/5-01 | M22520/5-02 Cavity B | M22520/5-01 | M22520/5-02 Cavity A | MS3323-16 | MS3342-16 |
| 121 | M22520/2-01 | M22520/2-25 | M22520/5-01 | M22520/5-04 Cavity A | M22520/5-01 M22520/10-01 | M22520/5-04 Cavity A M22520/10-04 Cavity A | MS3323-12 | MS3342-16 |
| 122 | M22520/2-01 | M22520/2-25 | M22520/5-01 | M22520/5-04 Cavity A | M22520/5-01 M22520/10-01 | M22520/5-04 Cavity A M22520/10-04 Cavity A | MS3323-12 | MS3342-16 |
| 123 | M22520/2-01 | M22520/2-25 | M22520/5-01 M22520/10-01 | M22520/5-04 Cavity B M22520/10-04 Cavity B | M22520/5-01 M22520/10-01 | M22520/5-04 Cavity A M22520/10-04 Cavity A | MS3323-12 | MS3342-16 |
| 124 | M22520/2-01 | M22520/2-25 | M22520/5-01 M22520/10-01 | M22520/5-04 Cavity B M22520/10-04 Cavity B | M22520/5-01 M22520/10-01 | M22520/5-04 Cavity A M22520/10-04 Cavity A | MS3323-12 | MS3342-16 |
| 125 | M22520/2-01 | M22520/2-25 | M22520/5-01 | M22520/5-04 Cavity A | M22520/5-01 M22520/10-01 | M22520/5-04 Cavity A M22520/10-04 Cavity A | MS3323-12 | MS3342-16 |





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TABLE IV Test limits.

| BIN code | Tensile strength (lbs) min | | Contact engaging and separation forces (ounces) | | | | Maximum contact resistance (millivolts) | |
|---|---|---|---|---|---|---|---|---|
| | | | Inner contact | | Contact assy | | | |
| | Inner contact | Outer contact | Min | Max | Min | Max | Inner contact | Outer contact |
| 120 | 6.0 | 15 | 0 5 | 7 0 | 2.0 | 20.0 | 8.0 | 4.0 |
| 121 | 6 0 | | | | | | | 2.0 |
| 122 | 16.0 | | | | | | | |
| 123 | 6 0 | | | | | | | |
| 124 | 32 0 | | | | | | | |
| 125 | 12 0 | | | | | | | |

Military part numbers. See table V.

TABLE V Military part number.

| BIN code | Military part number |
|---|---|
| 120 | M39029/6-01 |
| 121 | M39029/6-02 |
| 122 | M39029/6-03 |
| 123 | M39029/6-04 |
| 124 | M39029/6-05 |
| 125 | M39029/6-06 |

Custodians
Army - EL
Navy - AS
Air Force - 85

Review activities
Army - MU, MI
Navy - EC, SH, MC
Air Force - 11, 17, 99
DSA - ES

User activities
Army -
Navy - OS
Air Force -

Preparing activity
Navy - AS

Agent
DSA - ES

(Project 5935-N041-13)



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